Infrared Einstein Response from a Renormalized Spectral Entropy Functional

Jérôme Beau[1*]

[1*]Independent Researcher, France.

Corresponding author(s). E-mail(s): jerome.beau@cosmochrony.org;

Abstract

We investigate the covariant metric variation of a minimal spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ defined by a Laplace-type elliptic operator on a four-dimensional Riemannian manifold. Using heat-kernel methods and zeta regularization, we analyze the renormalized functional and classify the resulting geometric contributions by derivative order.

We show that the renormalized spectral stress tensor decomposes into a local two-derivative Einstein tensor term, a cosmological term, local four-derivative quadratic curvature contributions, non-local form factors, and a conformal anomaly piece. The Einstein tensor arises from the counterterm associated with the a_2 heat-kernel pole, in direct analogy with induced gravity in the sense of Sakharov.

In the weak-curvature regime where all dimensionless curvature invariants satisfy $R\,\ell_\chi^2 \ll 1$, the local two-derivative term dominates over higher-derivative and non-local contributions. The induced Newton coupling scales parametrically with the spectral cutoff as $G_N \sim 16\pi^2\ell_\chi^2$, up to scheme-dependent factors.

The result establishes that the infrared-dominant local part of the renormalized spectral entropy variation is proportional to the Einstein tensor, without assuming fundamental gravitational dynamics.

Keywords: Spectral geometry, Heat-kernel expansion, Induced gravity, Zeta regularization, Effective action, Einstein tensor, Renormalization

Contents

1 Introduction

Gravity is traditionally described as a geometric phenomenon. In general relativity, the gravitational interaction is encoded in the curvature of spacetime, and the Newtonian potential arises as a weak-field limit of the Einstein equations. This geometric formulation is remarkably successful. It nevertheless leaves open the conceptual question of whether gravitational dynamics must be fundamental, or whether it may arise as an effective response of a more primitive operator-based structure.

In parallel, operator-theoretic and spectral approaches have clarified how elliptic operators encode geometric and physical information. Functional determinants, resolvents, and heat-kernel asymptotics play a central role in quantum field theory, statistical mechanics, and spectral geometry [1, 2]. Variations of logarithmic determinants generate non-local responses through the inverse operator. This suggests a bridge between entropy-like functionals and long-range interaction.

In a previous analysis, we investigated a minimal mechanism by which a Newtonian interaction can emerge from the variation of a projective entropy functional in three spatial dimensions. For a positive Laplace-type operator A, the functional

$$S_\Pi[A] = \frac{1}{2} \log \det{}' A \tag{1}$$

has the weak-perturbation variation

$$\delta S_\Pi = \frac{1}{2} \operatorname{Tr}\big(A^{-1} \delta A\big) , \tag{2}$$

so that the Green kernel governs the spatial structure of the response. In three dimensions, the Green kernel of a Laplace-type operator exhibits a universal $1/r$ decay. Consequently, the entropy response inherits a Newtonian long-range profile without assuming a force law.

The goal of the present paper is to develop a covariant extension of this mechanism and to identify, in a controlled infrared regime, the leading local geometric content of the *renormalized* metric variation. We do not claim an exact equality between the metric variation of the bare determinant and Einstein's equations. We also do not discard non-local contributions, higher-derivative curvature terms, or the conformal anomaly. Instead, our central claim is the following.

Central claim (renormalized IR locality).

In four dimensions, for a minimal Laplace-type operator and in the weak-curvature regime where all dimensionless curvature invariants satisfy $R\,\ell_\chi^2 \ll 1$, the infrared-dominant *local* two-derivative part of the renormalized metric variation $\delta S_\Pi^{\text{ren}}/\delta g^{\mu\nu}$ is proportional to the Einstein tensor $G_{\mu\nu}$. The associated coefficient defines an induced Newton coupling whose scaling is set by the spectral cutoff scale ℓ_χ. The Einstein term arises from renormalization of the geometric sector, via the counterterm associated with the a_2 heat-kernel pole, in the standard sense of induced gravity.

Operationally, this result identifies a robust hierarchy. Local four-derivative contributions (quadratic curvature variations) and non-local form factors are suppressed in the infrared by additional powers of ℓ_χ^2, while the conformal anomaly remains as a distinct ultraviolet-sensitive contribution whose presence is tracked explicitly.

The remainder of the paper is organized as follows. In Section 2, we define the covariant projective entropy functional and set up the minimal Laplace-type operator choice. In Section 3, we summarize the heat-kernel expansion and the role of Seeley–DeWitt coefficients in the pole structure of zeta regularization. In Section 4, we define the renormalized spectral stress tensor and decompose it by derivative order into Einstein, quadratic-curvature, non-local, and anomaly pieces. In Section 4.5, we formulate the infrared hierarchy as a covariant derivative expansion and state the dominance criterion $R\,\ell_\chi^2 \ll 1$. In Section 5, we discuss the induced-gravity interpretation and the identification of the effective Newton coupling. We conclude in Section 6 with limitations and outlook.

2 Covariant projective entropy functional

2.1 Ellipticity and minimal Laplace-type operator

Let (M, g) be a smooth four-dimensional Riemannian manifold. We consider a second-order differential operator acting on scalar functions.

The analysis relies crucially on ellipticity. For a differential operator of order two, ellipticity means that its principal symbol is invertible away from the zero section.

We choose the minimal Laplace-type operator

$$A_g = -\nabla_g^2, \tag{3}$$

where ∇_g is the Levi-Civita connection. Its principal symbol is

$$\sigma_2(A_g)(x, k) = g^{\mu\nu}(x) k_\mu k_\nu. \tag{4}$$

On a Riemannian manifold, $g^{\mu\nu} k_\mu k_\nu > 0$ for all nonzero k. Therefore A_g is elliptic.

The principal symbol fully determines the local propagation structure. In particular, the metric enters exclusively through $\sigma_2(A_g)$. Lower-order terms do not affect ellipticity.

Ellipticity ensures:

- existence of the resolvent $(A_g + \lambda)^{-1}$ for $\lambda > 0$,
- discreteness of the spectrum on compact manifolds,
- local regularity of the Green kernel,
- validity of the short-time heat-kernel expansion,
- meromorphic continuation of the spectral zeta function.

More general scalar Laplace-type operators may include a curvature coupling,

$$A_g(\xi) = -\nabla_g^2 + \xi R. \tag{5}$$

The additional term ξR modifies only the zeroth-order endomorphism part of the operator and does not alter its principal symbol. It therefore does not affect ellipticity.

In this work we set $\xi = 0$. This is the minimal operator choice in which the effective geometry is entirely encoded in the principal symbol, without additional curvature input introduced at lower order.

The conformal value $\xi = 1/6$ cancels the a_2 coefficient in four dimensions and suppresses the induced Einstein–Hilbert counterterm. The present analysis focuses on the minimal elliptic case $\xi = 0$, for which the a_2 contribution is nonvanishing.

2.2 Regularized determinant and renormalization setup

We define the projective entropy functional by

$$S_\Pi[g] \equiv \frac{1}{2} \log \det{}' A_g, \tag{6}$$

where the prime denotes omission of zero modes, if any.

The determinant is understood in the zeta-regularized sense,

$$\log\det{}' A_g = -\zeta'_{A_g}(0), \tag{7}$$

with

$$\zeta_{A_g}(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{\,s-1}\, \mathrm{Tr}\big(e^{-tA_g}\big)\, dt, \tag{8}$$

initially defined for $\mathrm{Re}(s)$ sufficiently large and continued meromorphically to $s = 0$.

In four dimensions, the small-t expansion of the heat trace contains ultraviolet divergences. Renormalization is implemented by subtracting the pole terms associated with the heat-kernel coefficients and introducing a renormalization scale μ.

The resulting renormalized functional $S_\Pi^{\mathrm{ren}}[g;\mu]$ is finite but contains scheme-dependent local geometric counterterms. These counterterms are crucial for the emergence of the Einstein–Hilbert structure discussed below.

2.3 First variation and resolvent trace

For a perturbation $A_g \to A_g + \delta A_g$, the first variation of the determinant satisfies formally

$$\delta S_\Pi = \frac{1}{2}\, \mathrm{Tr}\big(A_g^{-1}\, \delta A_g\big)\,. \tag{9}$$

This identity shows that the response of the functional is governed by the resolvent kernel. In three spatial dimensions, this mechanism produces a universal $1/r$ Green profile, leading to a Newtonian response without assuming a fundamental force law.

In the present covariant four-dimensional setting, the metric variation defines a renormalized spectral stress tensor,

$$\mathcal{T}^{\Pi,\mathrm{ren}}_{\mu\nu} \equiv -\frac{2}{\sqrt{g}} \frac{\delta S_\Pi^{\mathrm{ren}}}{\delta g^{\mu\nu}}. \tag{10}$$

Its precise geometric decomposition depends on the heat-kernel structure of A_g. In particular, local two-derivative, local four-derivative, non-local, and anomaly contributions will appear with distinct origins. The classification of these terms and their infrared hierarchy is developed in the following sections.

3 Heat-kernel expansion and Seeley–DeWitt structure

3.1 Short-time heat-kernel expansion

For an elliptic Laplace-type operator A_g on a smooth four-dimensional Riemannian manifold, the heat trace admits a short-time asymptotic expansion as $t \to 0^+$ of the form

$$\mathrm{Tr}\big(e^{-tA_g}\big) \sim \frac{1}{(4\pi t)^2} \sum_{k\geq 0} t^k \int_M d^4x\, \sqrt{g}\, a_{2k}(x). \tag{11}$$

The coefficients $a_{2k}(x)$ are local curvature invariants constructed from the metric and its derivatives. They are the Seeley–DeWitt coefficients of the operator [1, 3, 4].

For the minimal scalar Laplacian ($\xi = 0$), with vanishing endomorphism $E = 0$ and bundle curvature $\Omega = 0$, the first coefficients are

$$a_0(x) = 1, \tag{12}$$

$$a_2(x) = \frac{1}{6}R(x), \tag{13}$$

$$a_4(x) = \frac{1}{(4\pi)^2}\frac{1}{360}\left(\frac{5}{3}R^2 - R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right). \tag{14}$$

Thus, in the decomposition

$$a_4 = \alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \tag{15}$$

the explicit numerical coefficients are

$$\alpha_1 = \frac{5}{3\cdot 360(4\pi)^2}, \quad \alpha_2 = -\frac{1}{360(4\pi)^2}, \quad \alpha_3 = \frac{1}{360(4\pi)^2}. \tag{16}$$

In four dimensions, the quadratic curvature invariants are not all independent. The Gauss–Bonnet density

$$\mathcal{G} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2 \tag{17}$$

is topological and does not contribute to the local equations of motion. The chosen basis $\{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ is therefore equivalent, up to a Gauss–Bonnet term, to the commonly used basis $\{R^2, R_{\mu\nu}R^{\mu\nu}, \mathcal{G}\}$.

3.2 Pole structure in four dimensions

Substituting the heat expansion into the definition of $\zeta_{A_g}(s)$ shows that in $d = 4$ the coefficients a_0 and a_2 generate simple poles at $s = 2$ and $s = 1$, respectively.

More precisely:

- a_0 controls the cosmological-constant counterterm,
- a_2 controls the Einstein–Hilbert counterterm,
- a_4 controls quadratic curvature counterterms.

After subtraction of the pole terms and introduction of the renormalization scale μ, the renormalized functional contains local geometric counterterms whose coefficients depend on μ.

3.3 Origin of the Einstein tensor

The crucial structural point is the following.

The coefficient $a_2(x) = R/6$ implies that the pole subtraction generates a local term proportional to

$$\int d^4x\,\sqrt{g}\,R. \tag{18}$$

Upon metric variation, this counterterm produces the Einstein tensor,

$$\frac{\delta}{\delta g^{\mu\nu}} \int d^4x \sqrt{g}\, R \ \propto\ G_{\mu\nu}. \tag{19}$$

Therefore, the local two-derivative Einstein term in the renormalized spectral stress tensor arises from the counterterm associated with the a_2 heat-kernel pole.

It does not arise from the bare finite part of $\zeta'_{A_g}(0)$ alone. Rather, it is generated through renormalization of the geometric sector, in the standard sense of induced gravity.

This mechanism is structurally identical to Sakharov's induced-gravity construction [5], with the spectral cutoff scale playing the role of the ultraviolet regulator.

Higher-derivative local contributions originate from a_4 and yield four-derivative tensors (Bach-like structures) upon variation. Non-local form factors arise from the finite part of the determinant and will be treated separately.

4 Metric variation and renormalized spectral stress tensor

4.1 Definition of the renormalized spectral stress tensor

The renormalized projective entropy functional $S_\Pi^{\mathrm{ren}}[g;\mu]$ defines a covariant stress tensor by

$$\mathcal{T}^{\Pi,\mathrm{ren}}_{\mu\nu} \equiv -\frac{2}{\sqrt{g}} \frac{\delta S_\Pi^{\mathrm{ren}}}{\delta g^{\mu\nu}}. \tag{20}$$

Formally, prior to renormalization, the first variation reads

$$\delta S_\Pi = \frac{1}{2}\,\mathrm{Tr}\big(A_g^{-1}\delta A_g\big)\,, \tag{21}$$

so that the response is governed by the resolvent kernel. After renormalization, local counterterms contribute additional geometric variations.

In four dimensions, and for the minimal Laplacian $\xi = 0$, the metric variation admits the general decomposition

$$\frac{\delta S_\Pi^{ren}}{\delta g^{\mu\nu}} = c_{\mathrm{E}H}(\mu)\, G_{\mu\nu} + c_\Lambda(\mu)\, g_{\mu\nu} + \beta(\mu)\, B_{\mu\nu} + T^{\text{non-}loc}_{\mu\nu} + A_{\mu\nu}. \tag{22}$$

Each term has a distinct geometric origin.

4.2 Local two-derivative sector

The Einstein tensor contribution arises from the renormalized Einstein–Hilbert counterterm generated by the a_2 pole. From Section 3, the subtraction of the a_2 divergence produces a local term proportional to

$$\int d^4x\,\sqrt{g}\,R. \tag{23}$$

Its metric variation yields

$$\frac{\delta}{\delta g^{\mu\nu}}\int d^4x\,\sqrt{g}\,R = -\sqrt{g}\,G_{\mu\nu}. \tag{24}$$

Therefore,

$$c_{\mathrm{E}H}(\mu) = \frac{1}{16\pi G_N(\mu)}, \tag{25}$$

where $G_N(\mu)$ is the induced Newton coupling.

The cosmological constant term originates from the a_0 pole and is proportional to $\int\sqrt{g}$.

4.3 Local four-derivative sector

The coefficient a_4 generates quadratic curvature counterterms of the form

$$\int d^4x\,\sqrt{g}\,\left(\alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right). \tag{26}$$

Their variation produces fourth-order differential tensors, which we collectively denote by $B_{\mu\nu}$. These are Bach-like structures containing four derivatives of the metric.

4.4 Non-local form factors and anomaly

The finite part of the determinant generates non-local contributions. In covariant form, these can be expressed through curvature-dependent form factors of the type

$$R\,\mathcal{F}\left(\frac{\Box}{\mu^2}\right)R, \tag{27}$$

and similar structures, as developed by Barvinsky and Vilkovisky [6, 7]. Their metric variation yields non-local tensors $T^{\text{non-loc}}_{\mu\nu}$.

In addition, renormalization induces a conformal trace anomaly. For a minimally coupled scalar field in four dimensions, the trace of the anomaly satisfies [1, 8]

$$g^{\mu\nu}A_{\mu\nu} = \frac{1}{(4\pi)^2}\frac{1}{360}\left(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}\Box R\right). \tag{28}$$

The local tensor structure of $A_{\mu\nu}$ depends on the renormalization scheme, but its trace is fixed by this universal expression. The anomaly contributes at four-derivative order and remains subleading in the weak-curvature infrared regime $R\,\ell_\chi^2 \ll 1$.

4.5 Infrared hierarchy

Consider the regime in which all dimensionless curvature invariants satisfy

$$R\,\ell_\chi^2 \ll 1, \qquad R_{\mu\nu}\,\ell_\chi^2 \ll 1. \tag{29}$$

In this regime, the derivative expansion is controlled by the small parameter $R\,\ell_\chi^2$. Since $B_{\mu\nu}$ is quadratic in curvature, its contribution scales as

$$\frac{\beta\,B_{\mu\nu}}{c_{\mathrm{EH}}\,G_{\mu\nu}} \sim R\,\ell_\chi^2. \tag{30}$$

Therefore, for $R\,\ell_\chi^2 \ll 1$, the local two-derivative Einstein term dominates over the four-derivative sector. Non-local form factors reduce to suppressed derivative corrections.

This establishes the infrared dominance of the Einstein tensor within the renormalized spectral stress tensor.

5 Identification of Newton's constant and spectral scaling

5.1 Running of the Einstein–Hilbert coefficient

In four dimensions, the subtraction of the a_2 pole generates a renormalization-scale dependence in the Einstein–Hilbert coefficient.

For the minimal scalar Laplacian with $a_2 = R/6$, the scale dependence of the induced coupling satisfies

$$\frac{d}{d\log\mu}\left(\frac{1}{16\pi G_N(\mu)}\right) = \frac{1}{(4\pi)^2}\frac{1}{6}. \tag{31}$$

This relation follows directly from the residue of the a_2 pole in the zeta-regularized determinant. It expresses the standard induced-gravity running generated by a minimal scalar degree of freedom [2, 8].

The absolute normalization of G_N depends on the ultraviolet completion or cutoff prescription. However, its parametric dependence is fixed by dimensional analysis.

5.2 Spectral cutoff and induced scale

Let Λ_{UV} denote the ultraviolet cutoff scale associated with the spectral regularization. Identifying Λ_{UV} with the inverse spectral resolution scale,

$$\Lambda_{\mathrm{UV}} \sim \ell_\chi^{-1}, \tag{32}$$

the induced Einstein–Hilbert coefficient scales as

$$\frac{1}{16\pi G_N} \sim \frac{\Lambda_{\mathrm{UV}}^2}{16\pi^2}. \tag{33}$$

Therefore,

$$G_N \sim 16\pi^2\, \ell_\chi^2, \tag{34}$$

up to scheme-dependent numerical factors and possible multiplicities from additional degrees of freedom.

This scaling relation provides the bridge between the spectral resolution scale ℓ_χ and the effective gravitational coupling.

5.3 Infrared consistency

The infrared hierarchy established in Section 4.5 requires

$$R\,\ell_\chi^2 \ll 1. \tag{35}$$

Using the scaling above, this condition is equivalent to

$$R \ll G_N^{-1}, \tag{36}$$

which is precisely the weak-curvature regime in which classical general relativity is valid.

Thus the dominance of the Einstein tensor in the renormalized spectral stress tensor is self-consistent in the classical infrared domain.

6 Discussion and outlook

6.1 Summary of the structural result

We have constructed a covariant projective entropy functional

$$S_\Pi[g] = \frac{1}{2}\log\det{}' A_g, \tag{37}$$

for a minimal elliptic Laplace-type operator.

Using heat-kernel methods and zeta regularization in four dimensions, we showed that the renormalized metric variation admits the decomposition

$$\frac{\delta S_\Pi^{ren}}{\delta g^{\mu\nu}} = c_{\mathrm{E}H} G_{\mu\nu} + c_\Lambda g_{\mu\nu} + \beta B_{\mu\nu} + T_{\mu\nu}^{\text{non-}loc} + A_{\mu\nu}. \tag{38}$$

The Einstein tensor arises from the counterterm associated with the a_2 heat-kernel pole. Local four-derivative terms originate from a_4. Non-local form factors and anomaly contributions are retained explicitly.

In the weak-curvature regime $R\,\ell_\chi^2 \ll 1$, the local two-derivative Einstein term dominates over higher-derivative contributions. The induced Newton coupling scales parametrically as

$$G_N \sim 16\pi^2 \ell_\chi^2, \tag{39}$$

up to scheme-dependent factors.

The result is therefore not an exact derivation of Einstein's equations from the bare determinant. It is a structural statement about the infrared-dominant local two-derivative part of the renormalized spectral stress tensor.

6.2 Relation to induced gravity

The mechanism identified here is structurally identical to induced gravity in the sense of Sakharov. The gravitational coupling emerges from renormalization of the geometric sector. The spectral cutoff scale sets the magnitude of the induced Einstein–Hilbert term.

No assumption of fundamental gravitational dynamics is made. The geometric response arises from operator-theoretic structure.

6.3 Limitations

Several limitations must be emphasized.

First, the analysis is performed in a Euclidean Riemannian setting. The extension to Lorentzian signature requires additional control over analytic continuation and causal structure.

Second, non-local contributions are not eliminated. They remain present and may become relevant outside the weak-curvature regime.

Third, the precise numerical value of the induced Newton constant depends on the ultraviolet completion and matter content.

Finally, this construction does not address dynamical stability, cosmological evolution, or coupling to additional fields.

6.4 Outlook

The present framework establishes that a minimal spectral entropy functional produces, after renormalization, an infrared-dominant local Einstein response.

Possible extensions include:

- inclusion of additional operator sectors,
- analysis of Lorentzian continuation,
- coupling to matter operators,
- exploration of regimes in which quadratic curvature terms dominate.

More generally, the result supports the view that geometric gravity may arise as an effective low-energy manifestation of spectral operator structure.

6.5 Remark on geometric flows and spectral monotonicity

The functional S_Π depends on the full eigenvalue spectrum of the elliptic operator A_g. Under geometric evolution equations such as the Ricci flow, spectral quantities associated with the Laplacian exhibit monotonic behavior.

In particular, monotonicity results for the first eigenvalue $\lambda_1(-\nabla_g^2)$ of the scalar Laplacian under Ricci flow have been established in the literature (see Perelman [9], and further developments in [10, 11]). These results indicate that geometric evolution can impose a controlled behavior on low-lying spectral data.

Although S_Π is not identical to Perelman's entropy functional, its spectral definition suggests a structural link between geometric flows and spectral entropy variation. Clarifying this relation would require a dedicated analysis and lies beyond the scope of the present work.

References

[1] Vassilevich, D.V.: Heat kernel expansion: user's manual. Phys. Rept. **388**, 279–360 (2003) https://doi.org/10.1016/j.physrep.2003.09.002

[2] Parker, L., Toms, D.: Quantum Field Theory in Curved Spacetime. Cambridge University Press, ??? (2009)

[3] Seeley, R.T.: Complex powers of an elliptic operator. Proc. Symp. Pure Math. **10**, 288–307 (1967)

[4] DeWitt, B.S.: Dynamical theory of groups and fields. Gordon and Breach (1965)

[5] Sakharov, A.D.: Vacuum quantum fluctuations in curved space and the theory of gravitation. Sov. Phys. Dokl. **12**, 1040 (1968)

[6] Barvinsky, A.O., Vilkovisky, G.A.: The generalized schwinger–dewitt technique. Phys. Rept. **119**, 1–74 (1985)

[7] Barvinsky, A.O., Vilkovisky, G.A.: Covariant perturbation theory. Nucl. Phys. B **333**, 471–511 (1990)

[8] Birrell, N.D., Davies, P.C.W.: Quantum Fields in Curved Space. Cambridge University Press, ??? (1982)

[9] Perelman, G.: The entropy formula for the ricci flow and its geometric applications. arXiv:math/0211159 (2002)

[10] Cao, X.-D.: Eigenvalues of $(-\delta + \frac{R}{2})$ on manifolds with nonnegative curvature operator. Math. Ann. **337**, 435–441 (2007)

[11] Ma, L.: Eigenvalue monotonicity for the ricci flow. Ann. Global Anal. Geom. **29**, 287–293 (2006)